



**Rajiv Gandhi Institute of
Petroleum Technology**
राजीव गांधी पेट्रोलियम प्रौद्योगिकी
संस्थान

Multi-Agent AI System for Risk-Aware, Explainable Hydrocarbon Exploration

Explainable, risk-aware AI for faster and smarter hydrocarbon exploration.

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The Challenge in Hydrocarbon Exploration

Complexity

- *Traditional exploration methods struggle with vast geological data and uncertain subsurface conditions.*
- *Current ML models are black-box, domain-specific, and lack uncertainty & ESG integration.*

Risk

- *High financial stakes demand accurate predictions to minimize costly drilling failures.*
- *Each exploration well can cost \$50 M – \$500 M, yet only ~30 % succeed.*

Speed

- *Industry needs faster decision-making to stay competitive in dynamic energy markets.*
- *Poor decisions lead to non-productive drilling, CO₂ footprint, and financial loss*

Impact:

- *Uncertainty & lack of interpretability*
- *Low trust from geoscientists.*
- *Regulatory pressure demands transparent, risk-audited AI recommendations.*

Why Multi-Agent AI?

Explainable Multi-Agent Intelligence for Smarter Drilling



Collaborative Intelligence

Multiple AI agents work together, each specializing in different aspects of exploration analysis.



Risk Assessment

Dedicated agents evaluate geological, financial, and operational risks in real-time.



Explainability

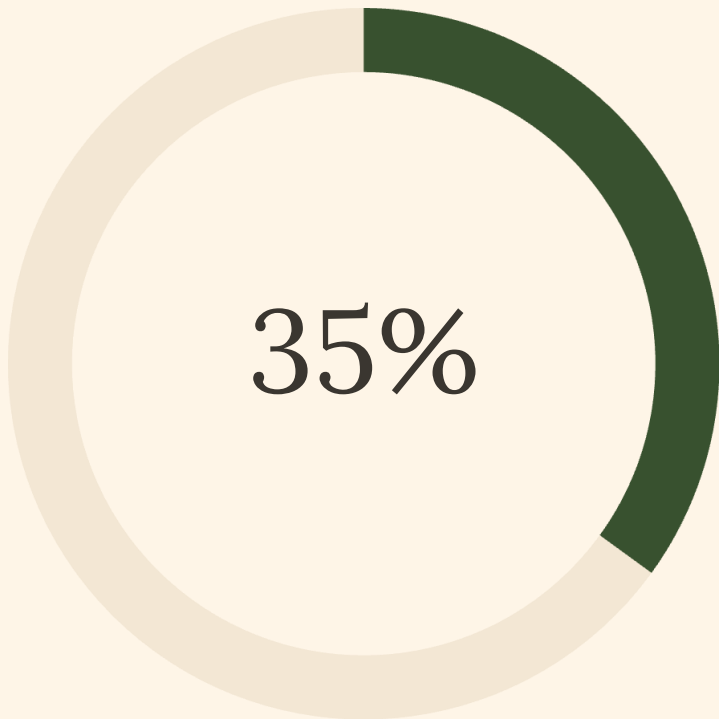
Transparent decision-making process shows how conclusions are reached, building trust.

Explainable Multi-Agent Intelligence for Smarter Drilling

- *A cutting-edge AI system that combines four specialized agents—**Seismic**, **Petrophysical**, **Reservoir**, and **Risk/ESG**—to deliver trustworthy, explainable drilling decisions.*
- *Each agent is trained on domain-specific data using advanced models including **CNNs**, **XGBoost**, and **Bayesian Monte Carlo simulations**.*
- *The system's **Consensus Engine** uses weighted voting, uncertainty fusion, and hard ESG veto logic to output ranked drill targets with confidence intervals, **SHAP explanations**, and **NPV forecasts**.*
- *This approach ensures **ESG-compliant**, financially optimized exploration that engineers and investors can trust.*

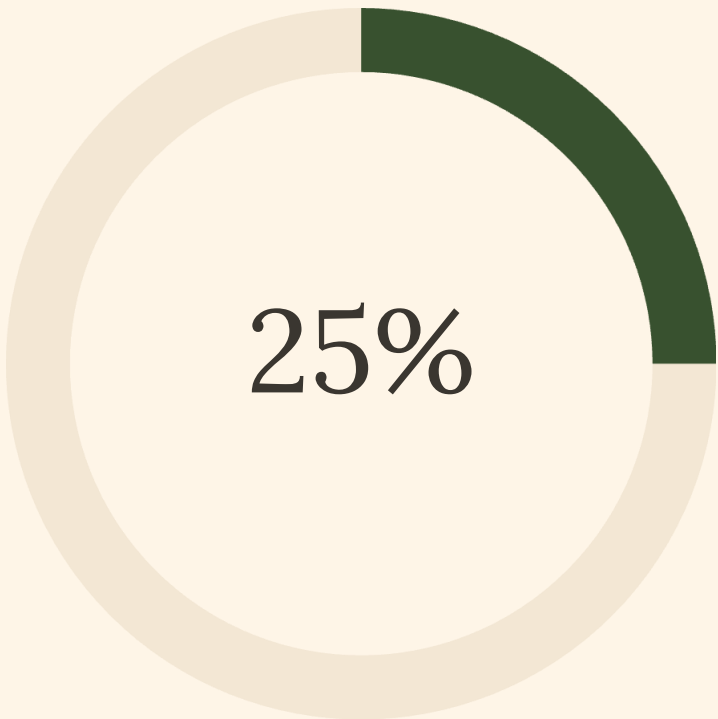


Risk-Aware Decision Making



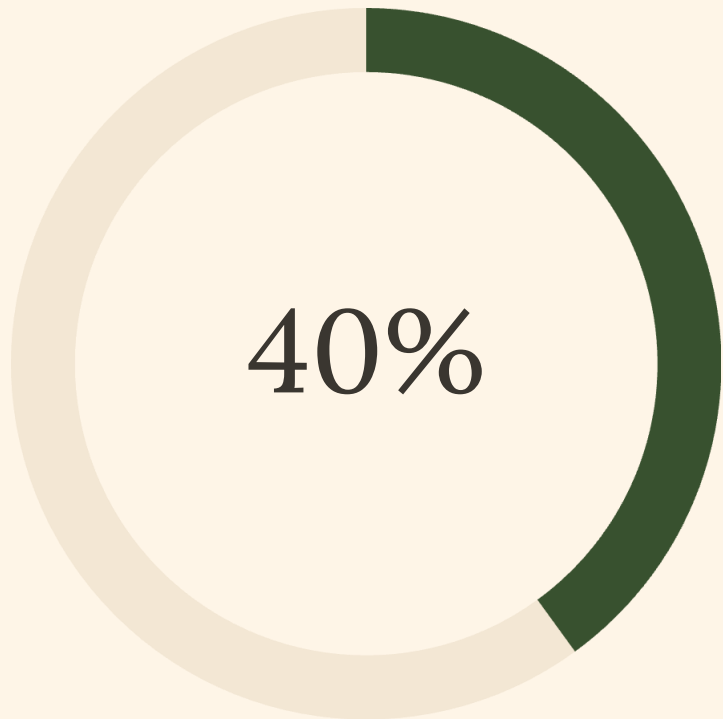
Geological Risk

Uncertainty in reservoir presence and quality



Technical Risk

Drilling and operational challenges



Economic Risk

Market conditions and project viability

Our system quantifies and visualizes multiple risk dimensions, enabling stakeholders to make informed trade-offs between potential rewards and uncertainties.

System Architecture



Data Ingestion Agent

Collects and preprocesses seismic, geological, and historical drilling data.



Analysis Agent

Applies machine learning models to identify potential hydrocarbon reservoirs.



Risk Evaluation Agent

Assesses probability of success and potential hazards for each prospect.



Explanation Agent

Generates human-readable reports detailing reasoning and recommendations.

System Architecture

Our multi-layered architecture processes diverse upstream data through specialized pipelines, delivering actionable intelligence with full explainability.

01 Input Layer <i>3D seismic data (SEG-Y format), FORCE 2020 well logs (LAS), production histories, economic parameters, and comprehensive ESG datasets feed the system.</i>	02 Processing Pipelines <i>Automated data ingestion, cleaning, normalization, and attribute extraction populate a centralized feature store for agent consumption.</i>
03 Agent Layer <i>Four specialized agents analyze data: Seismic Agent (CNN + Attention), Petrophysical Agent (XGBoost + CNN), Reservoir Agent (ResCNN + MLP), and Risk/ESG Agent (Bayesian Monte Carlo).</i>	04 Consensus Layer <i>Confidence-weighted fusion ($w = 1/\sigma^2$) integrates agent outputs, SHAP attribution explains feature importance, and ESG veto ensures compliance.</i>
05 LLM Layer <i>Structured prompts containing agent summaries, uncertainties, and geological maps enable reasoning, consistency checking, and human-readable report generation.</i>	06 Output Layer <i>Ranked drill targets, uncertainty bands, NPV distributions, and interactive dashboards with chatbot support deliver actionable recommendations.</i>

Inside the Agents

Agent	Input	Core Model	Output	Explainability
Seismic	3D SEG-Y	CNN + Self-Attention	Faults, Horizons, Probability Cubes	Grad-CAM maps
Petrophysical	Well Logs	XGBoost / CNN ensemble	Lithofacies (6 classes)	SHAP values
Reservoir	Merged features	Residual CNN + MLP	Reservoir Quality Index	Layer-wise Relevance
Risk/ESG	Economic + ESG data	Bayesian Monte Carlo	NPV Distribution, ESG Score	Rule explanation

Data Foundation & Implementation Scale

FORCE 2020 Dataset

Norwegian Continental Shelf data comprising 98 wells with approximately 75,000 analysis windows generated using 50-sample windows with 50% overlap for comprehensive coverage.

Log Curves (10 Total)

GR, NPHI, RHOB, DTC, DTS, RDEP, RMED, PEF, CALI, BS — covering the full spectrum of petrophysical measurements required for robust lithology identification.

Lithology Classes

- 1. Sandstone
- 2. Claystone
- 3. Chalk
- 4. Marl
- 5. Dolomite
- 6. Halite



LAS Parsing

95% success rate in automated data extraction and format validation



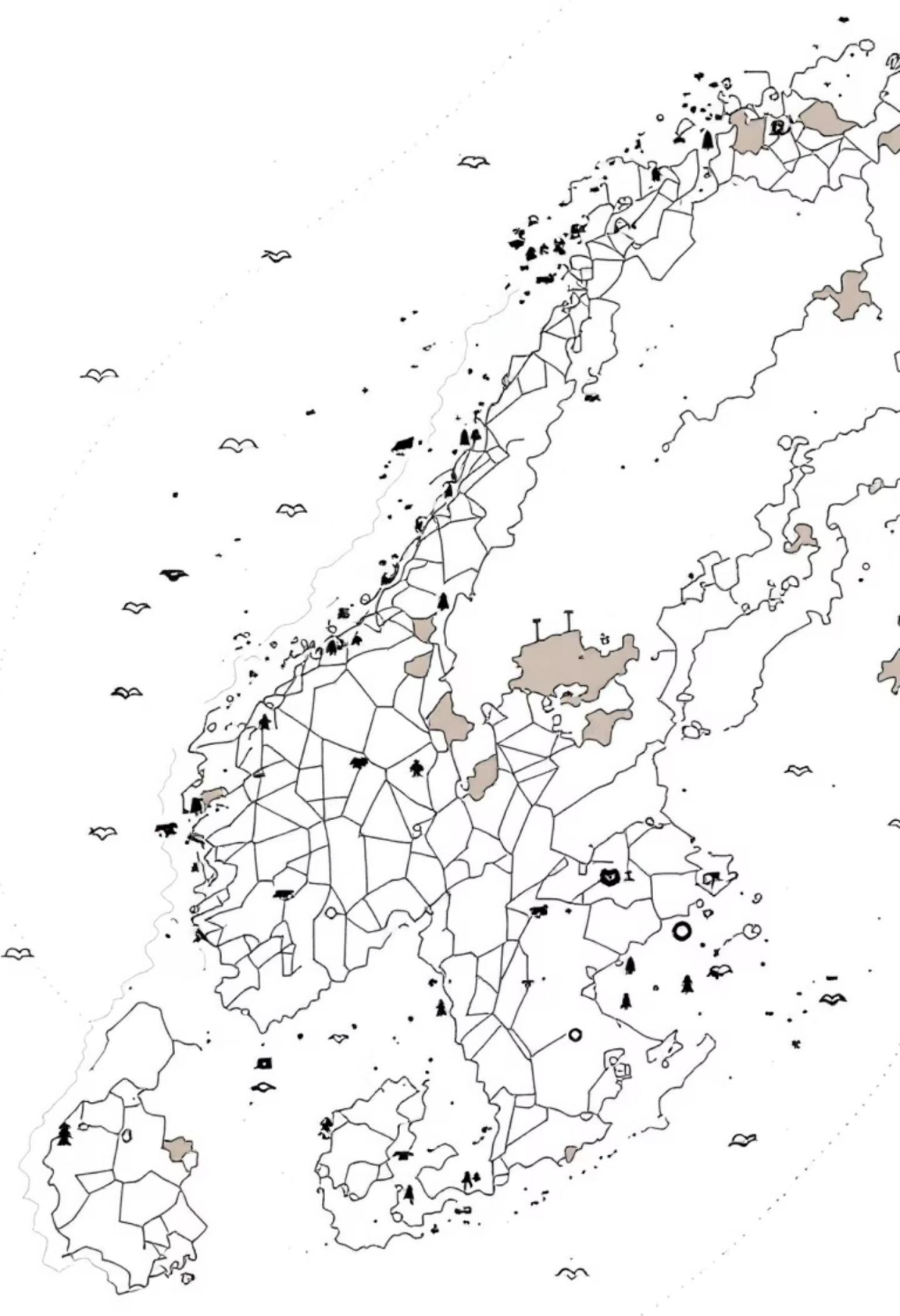
Sliding Windows

50 samples per window, 25-sample stride generating 70 features per window



Normalization

0–1 scaling with stratified split: 70% train, 15% validation, 15% test



Petrophysical Agent

The Petrophysical Agent combines deep learning with gradient-boosted trees to classify lithology from well log signatures with exceptional accuracy.

Model Architecture

1D CNN with attention mechanisms paired with XGBoost ensemble

Performance Metrics

- *Overall Accuracy: 85.2%*
- *F1 Score: 0.84*
- *Model Parameters: ~2.1M*

Top Classifications

- *Sandstone: 89% accuracy*
- *Halite: 91% accuracy*
- *Claystone: 82% accuracy*



Baseline Results:

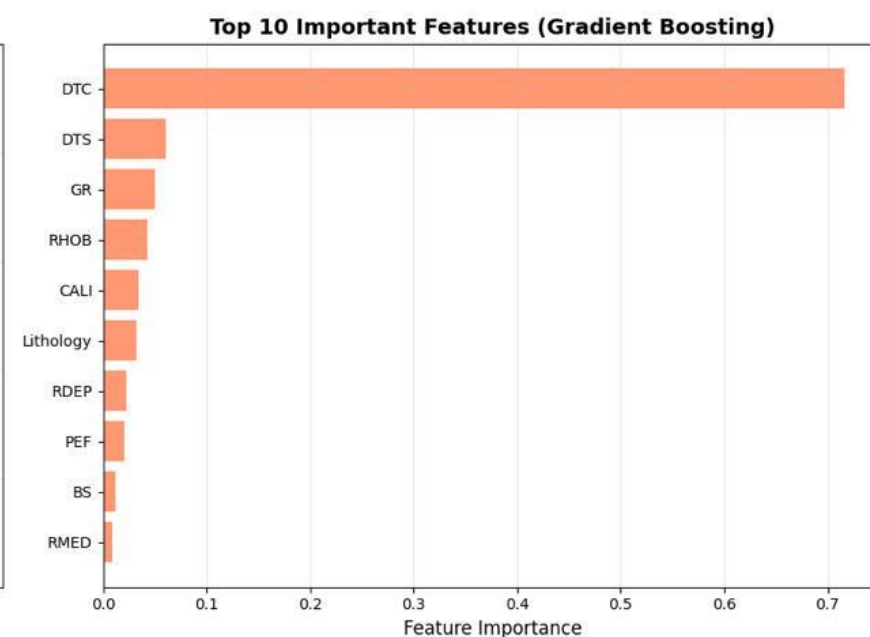
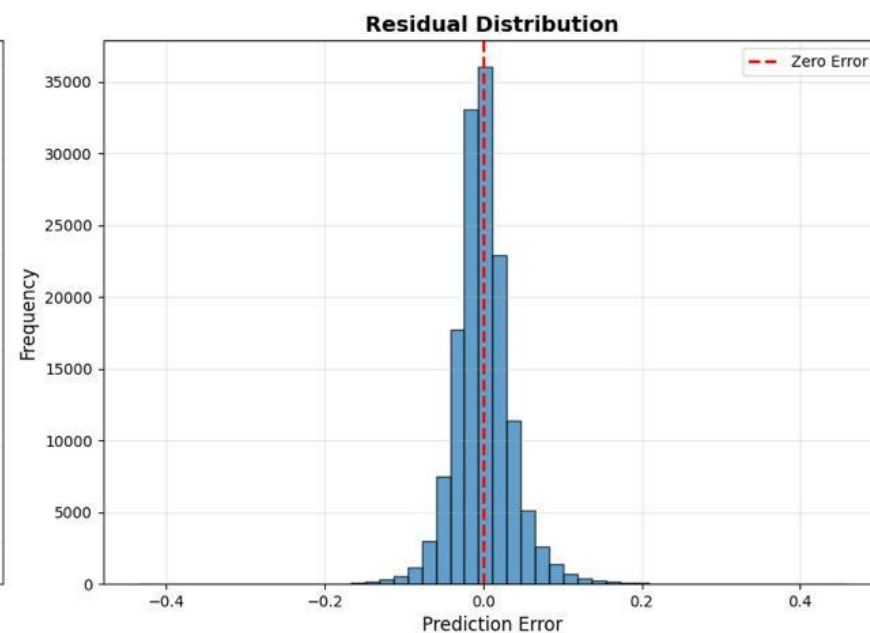
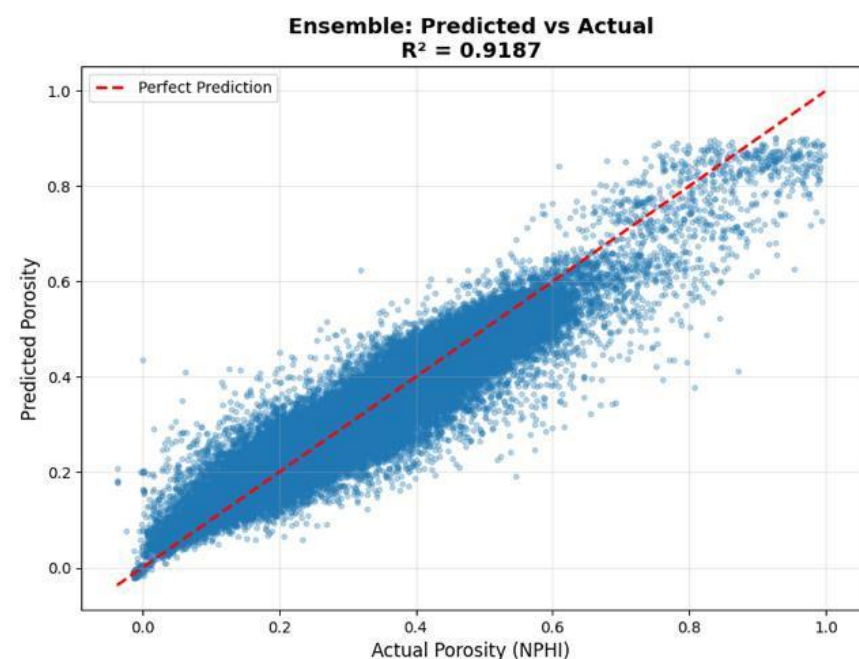
Training Accuracy: 98.17%

Validation Accuracy: 82.14%

Test Accuracy: 82.47%

The model demonstrates robust performance across diverse lithological classes, with MC Dropout (20 iterations) providing reliable uncertainty quantification for downstream decision-making.

Reservoir Agent



Model: Residual CNN + MLP
Target: Porosity & reservoir quality index
Metrics:

- $R^2 = 0.91$
- **RMSE = 0.028**

Individual Model Performance (Test Set):

Random Forest R^2 : 0.9339

Gradient Boosting R^2 : 0.8784

Neural Network R^2 : 0.9000

Ensemble R^2 : 0.9187 ★



Initial Results

Early validation demonstrates strong predictive performance across both classification and regression tasks. These models are under active development as we refine the complete end-to-end workflow.

Petrophysical Agent

Lithology Classification

- *Accuracy: 85.2% (↑3.2% from baseline RF)*
- *F1 Score: 0.84*
- *Model Size: ~2.1M parameters*

Per-Class Performance: Sand 89%, Halite 91%, Dolomite 86%, Clay 82%, Chalk 78%, Marl 81%

MC Dropout with 20 iterations provides robust uncertainty estimates.

Reservoir Agent

NPHI Regression

- *R^2 Score: 0.91 (validation: 0.92)*
- *RMSE: 0.028*
- *MAE: 0.021*

Feature Importance: Bulk Density (32%), Lithology (18%), GR (14%)

Ensemble architecture delivers 3–5% performance boost over single models, with exceptional generalization to held-out wells.

Sample Output



Well X Depth- 2450 m

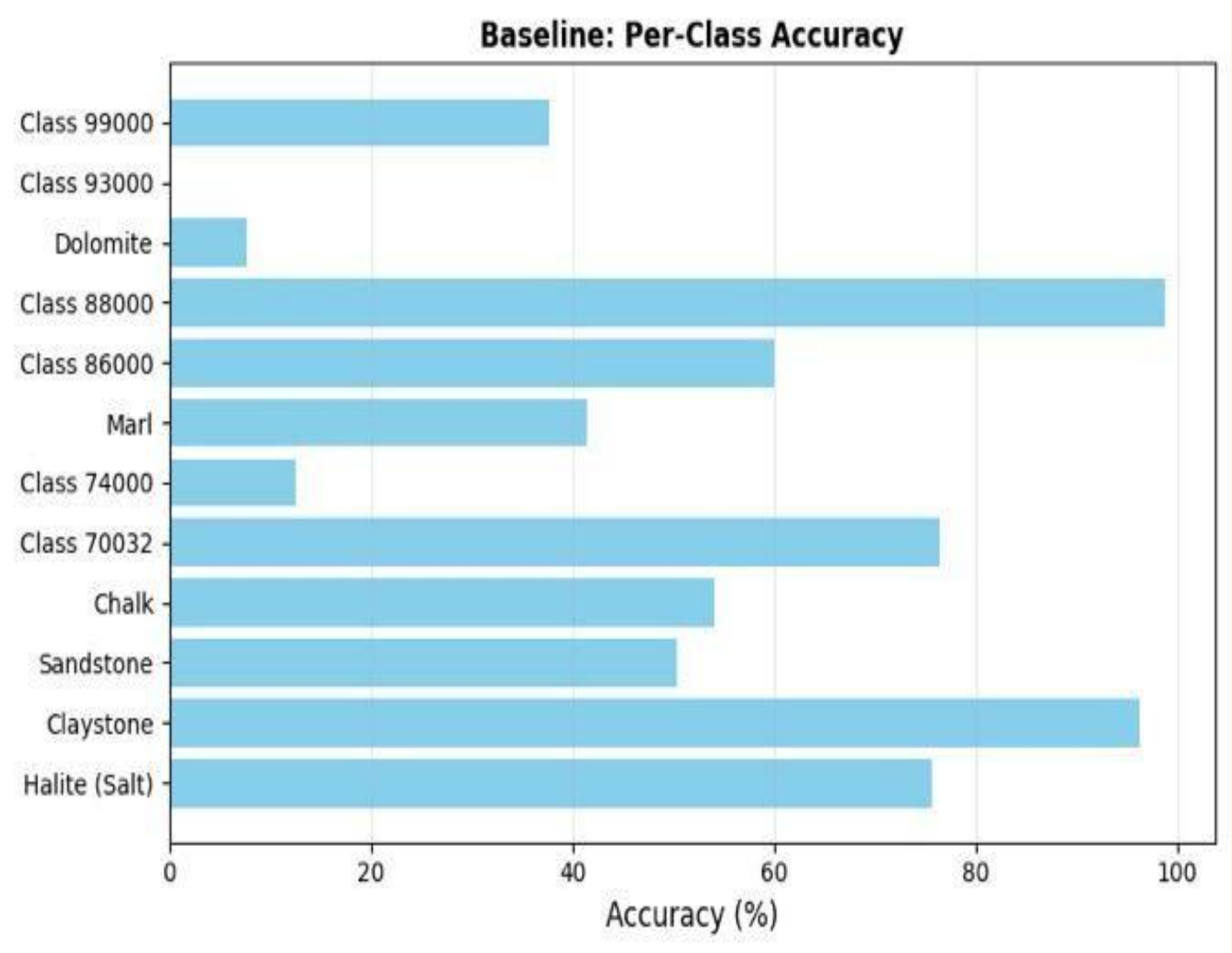
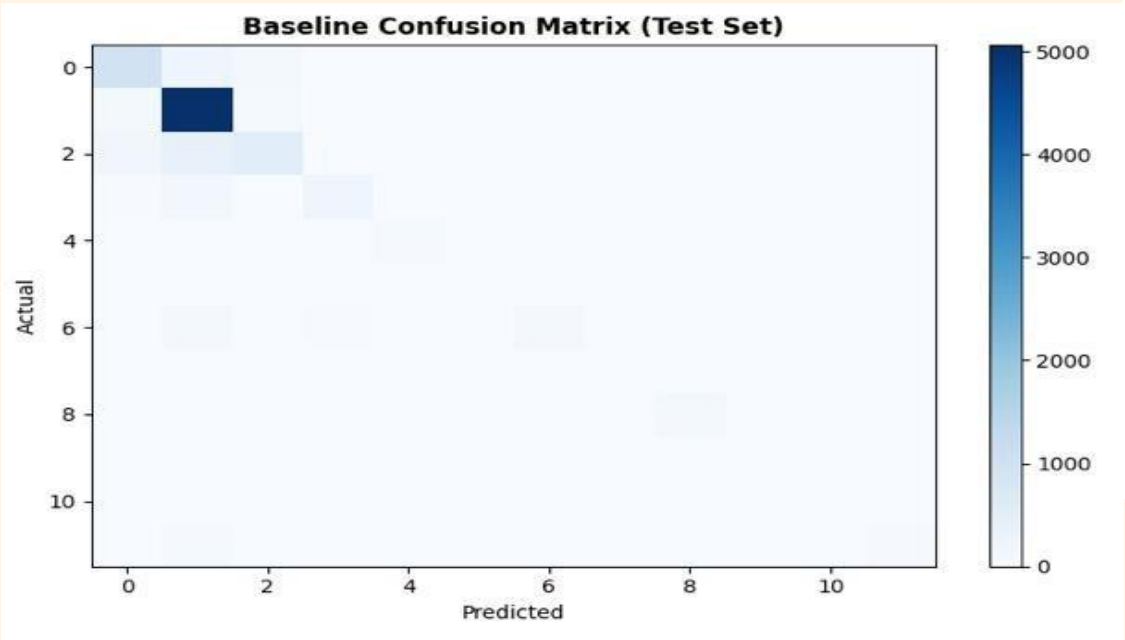


Lithology: Sandstone (91% confidence)

Porosity: 18.2%

NPV (P50): \$45M

ESG: PASS → Recommend DRILL





Expected Impact

40%

Faster Analysis

*Reduced time from
data collection to
drilling decision*

30%

Cost Reduction

*Fewer dry wells
through improved
targeting accuracy*

50%

Success Rate

*Increased probability of
discovering viable
reserves*

Technology Stack



Machine Learning

Deep learning models for seismic interpretation and pattern recognition in geological data.



Cloud Computing

Scalable infrastructure for processing massive datasets and running complex simulations.



Visualization Tools

Interactive dashboards for exploring results and understanding AI recommendations.

Roadmap to Implementation

Phase 1: Development

Build and train individual AI agents using historical exploration data.

Phase 3: Validation

Test system on real-world cases and refine based on expert feedback.

1

2

3

4

Phase 2: Integration

Connect agents into unified system and develop explanation framework.

Phase 4: Deployment

Roll out to exploration teams and provide training for effective use.



Transforming Hydrocarbon Exploration

Our multi-agent AI system combines cutting-edge technology with industry expertise to make hydrocarbon exploration faster, smarter, and more transparent.



Precision targeting of drilling locations



Comprehensive risk assessment



Full transparency in decision-making